
EcoAI cooling

Making Data Centers Water-Neutral

AI-powered
intelligent cooling

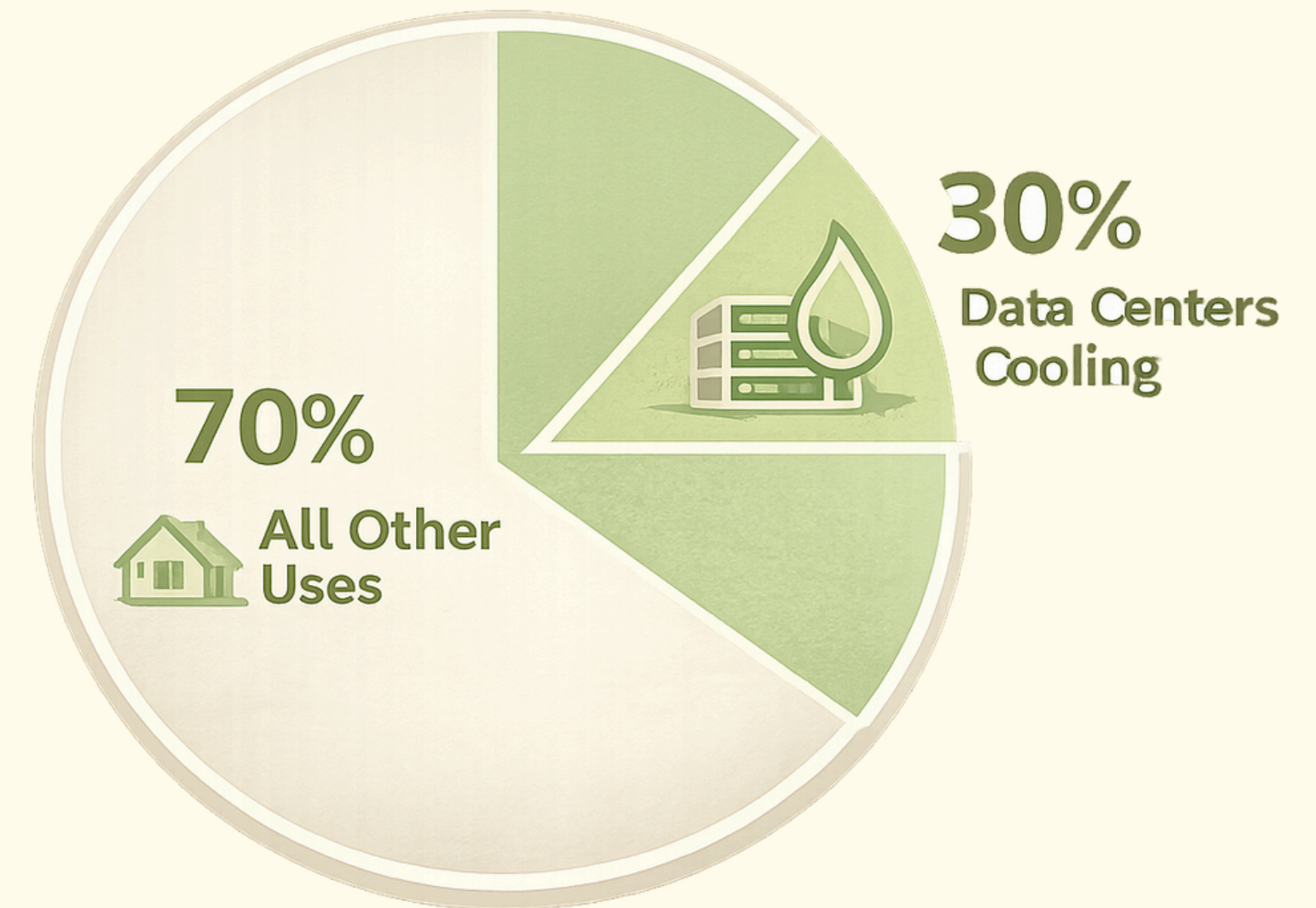


closed loop water
recycling for
sustainable digital
infrastructure



What is the problem?

- Data centers use billions of liters of water every year for cooling.
- One large facility can consume millions of liters per day.
- Much of this water is lost through evaporation.
- As AI grows, water demand grows.



Water Usage in a High
Data Center Area

Problem & Impact



Who is affected?

- Local communities (water scarcity)
- The environment and ecosystems
- Wildlife in polar regions
- Data center operators (high costs & regulations)
- Future generations

Why does it matter?

- Increased carbon emissions
- Water scarcity
- Global warming
- Melting ice caps & rising sea levels
- Environmental pollution
- Energy and water waste
- Negative impact on local communities

Our Solution

EcoAI Cooling

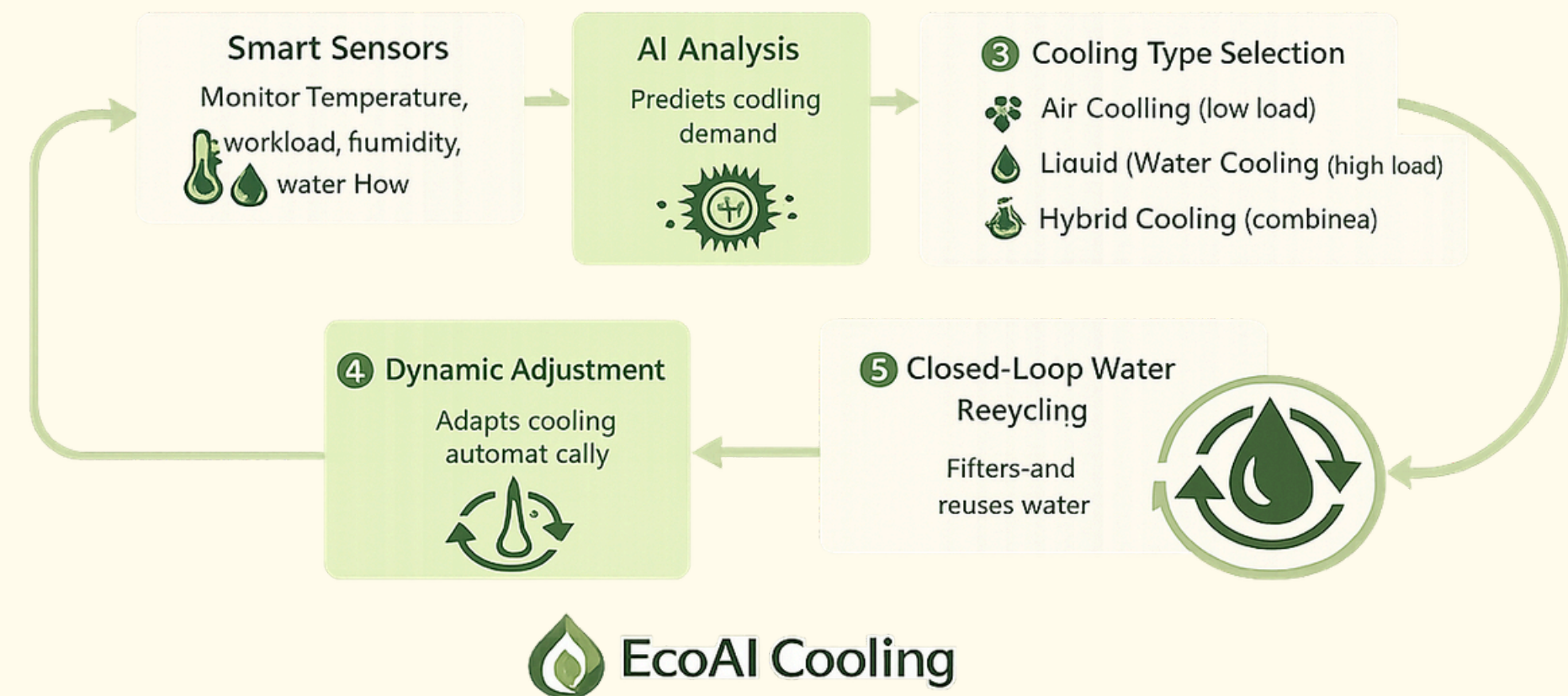
EcoAI Cooling is an intelligent cooling management system for data centers that reduces water waste and energy overuse.

EcoAI Cooling uses sensors and AI to:

- Predict cooling demand
- Select the most efficient cooling type (air, liquid, hybrid)
- Adjust cooling automatically
- Recycle water in a closed-loop system

Smarter cooling. Less water. Lower impact.

EcoAI Cooling – How It Works



Who Are Our Competitors?

- Competitors focus on single solutions.
- We combine AI and water recycling.
- This gives a smarter, integrated system.

Most of them solve one part of the problem.

EcoAI Cooling	Traditional Cooling
Predictive cooling	Fixed cooling load
Closed-loop reuse	High water loss
AI optimization	Reactive systems
Software-driven	Hardware-focused

Solution

We Combine What Others Separate

Most companies focus on:

- Cooling hardware only
- OR energy monitoring software
- OR water-saving systems

EcoAI Cooling integrates all three in one intelligent platform.

Why Our Solution?

We don't replace infrastructure.

We optimize existing systems using sensors + AI software.

Lower cost. Faster adoption. Scalable.

Smart Decision Making, Not Just Cooling

Traditional systems cool continuously.

Our AI:

- Predicts demand
- Selects the best cooling type (air, liquid, hybrid)
- Prevents overcooling before it happens

This reduces both energy and water waste.

Built-In Water Recycling

Most data centers still rely on fresh water supply.

EcoAI Cooling includes a closed-loop water reuse system, reducing dependency on new water sources.

Roadmap & Milestones

Build first prototype, hire 2 software developers, partner with hardware company, start pilot data center.

Budget: \$300,000

Prototype & Seed Funding Round



Deploy water recycling hardware, expand to multiple data centers, optimize system performance.

Budget: \$200,000

Hardware Expansion

Large scale deployment, enter new markets, scale AI and hardware solutions globally.

Budget: \$100,000

Full Commercial Launch & Global Expansion

AI Development & Pilot Expansion

Develop AI system, integrate sensors, expand pilot testing to more data centers.

Budget: \$250,000

Partnerships & Market Expansion

Improve efficiency, form partnerships with data center operators, prepare for commercialization.

Budget: \$150,000

Sustainable Development Goals

SDG 6 – Clean Water & Sanitation

- Reduces water waste in data centers
- Promotes responsible water reuse
- Supports water efficiency in industrial systems

SDG 9 – Industry, Innovation & Infrastructure

- Improves efficiency of digital infrastructure
- Integrates AI into sustainable engineering systems

SDG 12 – Responsible Consumption & Production

- Reduces resource waste
- Optimizes operational efficiency
- Encourages smarter industrial resource management

SDG 13 – Climate Action

- Lowers environmental footprint of digital growth
- Supports climate-resilient infrastructure



Impact

How We Measure Impact (KPIs)?

- **Water Reduction Rate**

We aim to reduce water consumption by 30–50% per data center.

- **Water Reuse Efficiency**

Achieve 80–90% water reuse through a closed-loop cooling system.

- **Energy Optimization**

Lower cooling energy usage by 15–25% using AI-based optimization.

- **CO₂ Emission Reduction**

Measure the annual reduction in carbon emissions from cooling operations.



Meet Our Team



**Zakia Mahmoud
Hanafi
CEO & Founder**

Software Engineer & AI Track
Student
Focused on AI-driven solutions,
system design, and data analysis for
sustainable technology.



**Alradia SaifEldin Osman
Mohamed Hamed
Co-Founder & Hardware Lead**

Civil Engineer
Specialized in infrastructure and
physical system feasibility for
sustainable cooling solutions.

EcoAI cooling

Thank You